This project is to be done in groups of two. You can form your own group. This is an extension of the CPU project that was done as part of the Digital Systems course (EE224). The goal is to implement a pipelined version of the CPU. It will be a 8-bit CPU with 8 registers. The pipelined datapath implementation will be restricted to just ALU (Arithmetic) operations. You do not have to worry about pipeline hazards (to begin with).

As some groups did not complete the project, you can use one of the baselines provided (shared by your classmates). This is provided so that you can focus on the changes required to implement the pipelined datapath.

Two possible baselines are shared here. If you know of other batches (from your class) whose code can be used as baseline, you can do so. But acknowledge the source in your final report.

- Manan Garg and team: `https://github.com/MananGarg-iitb/IITB_CPU`
- Daksh Sawke and team: `https://github.com/amrutanshu-mohanty/EE-224-IITB-CPU`

Possible steps to be taken for the project completion.

- Verify that the baseline VHDL (or verilog) code for the CPU functions correctly. Do few tests and ensure this. Make sure you can run atleast 5 instructions sequentially.
- Modify the baseline so that it only has the datapath that supports arithmetic operations. Verify your modification works correctly. Note the time taken for five sequential instructions.
- Modify the above datapath to be a pipelined version with five pipeline stages. Verify the functionality using a single instruction. Next, verify it by executing five instructions sequentially. Compare the time required for the five instructions with the time taken in the baseline version.
- Make a report of the modifications that were done to obtain the pipelined datapath.

Demonstrate this to the TAs.

1. Demonstrate your baseline model by running five sequential instructions.
2. Demonstrate your pipelined version by executing a single instruction.
3. Demonstrate your pipelined version by executing five sequential instructions.
4. Submit your project report on moodle.